

Let's Talk About MATH

More importantly, the linguistic barrier blocking the everyday person in believing in their mathematical capabilities, which blocks them from the most fast-growing fields.

If I ask...

"Who here thinks they're bad at math?" most in this room would probably nod and agree that they aren't the best but at the end of this presentation I will show you how you already use advanced mathematical concepts in your everyday life without your knowledge.

But First, Let's Examine Why You THINK You're "Bad" At Math

64% of people report having math anxiety.

And the effects are HUGE.

As mentioned, this barrier tracks into a path which locks you out of the fastest-growing fields, which was proven by Özdemir [Urz-deh-meer] in 2025.

When your body experiences math anxiety, its stress response mimics is the same as if you experienced a physical threat, which causes a feedback loop, documented by Ashcraft and Krause [KROW-zuh]. Lyons and Beilock further studied this using fMRI scans in 2012.

It's a Language Barrier

There are countless of studies that support this reality.

When students, professors, everyone in a professional field who claims they are bad at math, they are really saying that the vocabulary and notation is what sets them apart from those who have mastered the language.

Scary Terms You Already Understand

Take a look at these. And, as the examples appear, I'd like you to raise your hand high and proud when you see one you understand.

Let's get started.

Taylor Series

It's applied in your everyday life when your GPS estimates your ETA.

Graph Theory

You apply this whenever you work on an assignment that has multiple steps, such as the speeches in this class. You must choose a topic first, do research, and so forth. You know that you can't practice

your speech if you don't even have a topic.

Epsilon-Delta Limits

Also get applied whenever you tell your friend, "I'm almost there" even though you haven't left the door.

Markov Chains

When you're typing on your phone and it tries to predict the next word.

The Barrier Isn't Ability But How Math is Presented

If we focus on the concepts at the beginning and introduce the fancy language and notation afterwards gradually, students would feel more confident in their capabilities.

Ethnomathematics Framework

This is further proven through this framework first introduced in 1985. Here are some examples people have studied over time but you also experience this yourselves.

You are able to calculate discount, tax, and total cost at your favorite store and what your total cost will be in just a few seconds.

Yet when it comes to being given a test on these same concepts, the processes seem to have vanished from your mind. Weird, right?

Math You Already Use Daily

Here are some more examples of what YOU call advanced math that you perform on a regular basis.

Again, raise your hand if you personally do or experience one of these on a regular basis.

Real Analysis

You adjust your thermostat all the time, especially these days with the weather not being able to make up its mind whether it wants to be hot or cold.

Linear Optimization

When shopping, you always (hopefully) want to save your hard earned money and compare unit prices to find the best deals.

Chaos Theory

You know if you wake up just ten minutes late, it might end up with you leaving the house later and hitting traffic you usually avoid getting to work or school, which leads to a whole bunch of unneeded stress.

Knot Theory

You apply this, or used to before everything became wireless - so maybe your generation haven't even experienced this struggle, when you untangled your headphones you stuffed in your pockets or in your bag.

Set Theory

Going back to shopping, when you use those filters. You're applying set theory.

Universal Quantifiers

You do this nearly everyday when you say things like "Everyone failed that chemistry test" or when your professor says "Everyone in this class will get 20pts extra credit". Yeah? Just a suggestion.

Ready For Some Math?

Don't worry you won't have to do any.

But I wanted to share some problems math that I thought you might find interesting.

Deck of Cards

First off, I'll have to thank Ray for bringing this one up. He mentioned that when you shuffle a deck of cards, there's a possibility that no one else in history has ever had the same deck that you're holding.

And the math proves it. The possible arrangements of a shuffled deck of cards is in fact significantly larger than the number of atoms on Earth.

Jackpot

The classic jackpot case.

But what I thought would be funny for you guys to see are the odds listed at the bottom of this table. Take a look.

Password Creation

You know how those websites keep telling you to change your password every 3-6 months? But who is going to remember all those passwords?

Here are the possibilities of how many passwords you could make with the different requirements websites have.

Conclusion

In the end, you aren't "bad at math".

You use advanced mathematics every single day.

You simply do not understand the vocabulary and notation, which should be approached as if you are learning a whole new language.

So, stop saying "I'm bad at math".

Instead say, "I'm learning the language".

Thank you.

If you are interested in knowing more about my research behind this topic, I promise I have about 70 pages on this and 40 pages of the math behind some of the examples you just saw.

